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Progress Review Report

Abstract
CurrencyGuard • is an AI-based system to detect counterfeit Indian Rupee notes using CNN, SVM, and Random Forest models. Built with FastAPI, TensorFlow, and SQLite, it provides real-time currency authentication through a full-stack web application.

1. **Introduction**
Counterfeit INR poses serious economic threats. Manual verification is slow and error-prone. CurrencyGuard • automates detection for •10–•2000 denominations using three ML models: CNN, SVC, and Random Forest.

2. **Problem Statement**
Manual currency inspection is slow, inconsistent, and hard to scale. Existing ML solutions don't target INR-specific security features. An automated, INR-focused detection system is needed.

3. **Dataset & Preprocessing**
8,760 images across 7 denominations from Kaggle. Split: 70% train, 15% validation, 15% test. 71.4% genuine, 28.6% counterfeit. Images resized to 64×64, grayscale, normalized, and augmented.

4. **Models**

CNN — 3 conv blocks, BatchNorm, Dropout, trained with Adam
optimizer SVC — RBF kernel on extracted security features
Random Forest — Ensemble classifier on tabular features

5. **Results**
ModelAccuracyPrecisionRecallF1CNN87.4%74.0%87.5%80.1%RF86.4%78.3%68.4%73.0%SVC85.8
%74.9%70.9
%72.9%
CNN achieved the best recall — critical for minimizing missed counterfeits.

6. **System Architecture**
Frontend: HTML/CSS/JS with upload, manual input, and dashboard. Backend: FastAPI + SQLite. Models: TensorFlow (CNN), Scikit-learn (SVC, RF). Includes Grad-CAM for explainability.

7. **Conclusion**
CNN outperformed other models with 87.4% accuracy and 87.5% recall. The system delivers real-time, explainable, multi-denomination counterfeit detection ready for deployment.

8. **Future Scope**
Transfer learning (ResNet/EfficientNet), TensorFlow Lite mobile app, live camera scanning, blockchain authentication, RBI deployment.

Signature of Guide: